Atlanta, US **CARLOS MARÍ NOGUERA** carlosmarinoguera@gmail.com

Education:

Georgia Institute of Technology
Master's degree in Electrical and Computer Engineering
Thesis Option: The use of Federated Learning in Seismic Interpretation
Advisor: Dr. Ghassan AlRegib
08/2025 – 12/2026 GPA 4.0/4.0

Universidad Pontificia de Comillas, ICAI
Master's degree in Telecommunication Engineering (MSc)
2024 - 2026 GPA: 8.9/10 (Spain)

Universidad Pontificia de Comillas, ICAI
Bachelor's degree in engineering in Telecommunications Technologies
2020 - 2024 GPA: 7.94/10 (Spain)

The University of Texas at Austin
Exchange: Department of Electrical and Computer Engineering
2023 – 2024 GPA: 3.85
Senior Design Project Mentor: Dr. Gustavo de Veciana

Coursework: Algorithms, Microprocessors, Digital Image Processing, Computer Vision, Microelectronics, Optoelectronics, Advanced Wireless Networks.

Experience:

05/26 – 08/26 Invopop (YC 23) Madrid, Spain
Product Engineer Intern
- Led the integration for the Danish domestic e-invoicing network.
- Developed internal tools to expedite client support using Go and Python cutting average client ticket resolution time.

08/24 – 12/25 Instituto de Investigación Tecnológica Madrid, Spain
Research Assistant
- Researched the use of CVAEs on biological timeseries compression and generation.

06/23 – 08/23 Tower Consultores Madrid, Spain
Business Intelligence Intern
- Built reports based on financial and performance data to aid management.

09/22 – 05/23 Universidad Pontificia de Comillas, ICAI Madrid, Spain
Collaborator Student
- Managed Structured Databases
- Conducted a cybersecurity audit on the university international website.

Skills: Programming Languages: Java, Python, C, C++, Go, JavaScript, MATLAB.
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, ONNX
Languages: English, Spanish, Catalan.

Research & Projects:

08/24 – 12/25 Use of Autoencoders in Time series forecasting Advisor: Dr. Mario Castro
08/23 – 06/24 Network optimization via RL. Advisor: Dr. Amy Zhang
10/23 – 12/23 Diffusion Model for Movie Poster Generation
- Engineered a custom diffusion model on PyTorch.

Open-source contributions:

02/24 Contributions to Flower Framework.
04/26 – 08/26 Contribution to GOBL

Publications:

1. C. Marí, Z. Fowler, G. AlRegib. "The Effectiveness of Federated Learning in Seismic Interpretation", International Meeting for Applied Geoscience & Energy (IMAGE), 2026.